

Relative Ease of Hydrolysis (Cambridge (CIE) A Level Chemistry): Revision Note

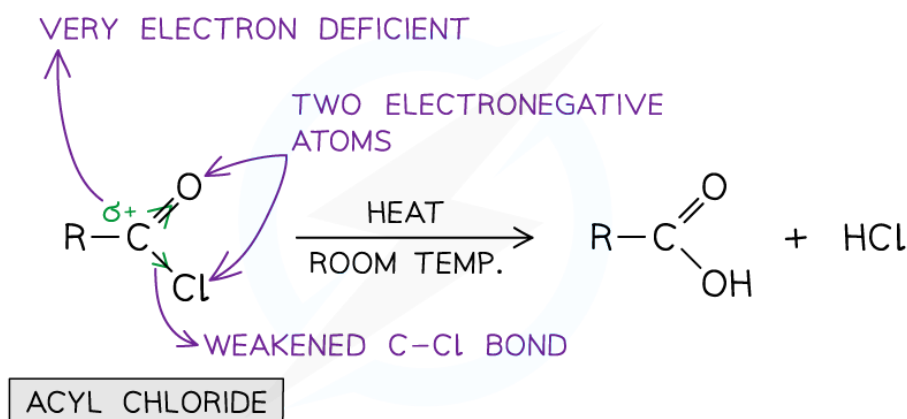
Hydrolysis of Acyl Chlorides, Alkyl Chlorides & Halogenoarenes

- **Hydrolysis** is the breakdown of a compound using **water**
- The ease of hydrolysis for different organic compounds may differ
- For example, the ease of hydrolysis, starting with the compounds most readily broken down, is: acyl chloride > alkyl chloride > aryl chloride
- This trend can be explained by looking at the **strength** of the C-Cl

Strength of C-Cl bond in acyl chlorides

- Acyl chlorides are hydrolysed most readily at **room temperature**
- This is because the carbon bonded to the chlorine atom is also attached to an oxygen atom
- There are two **strong electronegative** atoms pulling electrons away from the carbonyl carbon, leaving it very δ^+
- The C-Cl bond is therefore **weakened** and **nucleophilic attack** of the carbonyl carbon is much more **rapid**

Hydrolysis of acyl chlorides

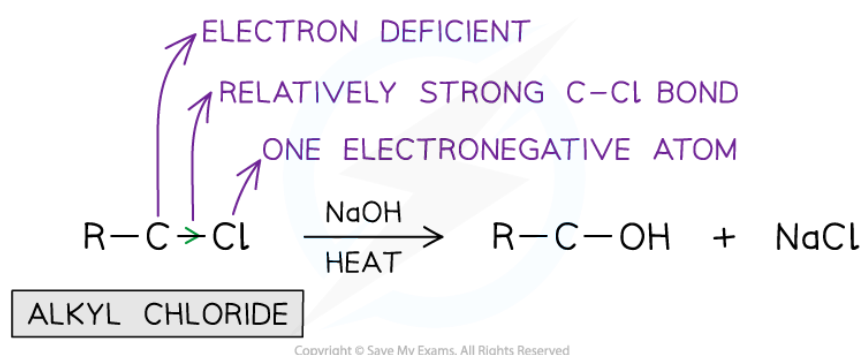


The hydrolysis of acyl chlorides occurs most readily

Strength of C-Cl bond in alkyl chlorides

- The carbonyl carbon in alkyl chlorides is only attached to **one electronegative** atom which pulls electrons away from it
- This carbon atom is therefore not very δ^+ and the C-Cl bond is stronger than the C-Cl bond in acyl chlorides
- The hydrolysis of alkyl chlorides, therefore, requires a **strong alkali** (such as OH^-) to be **refluxed** with it
- An OH^- ion will hydrolyse the alkyl chloride as it is a **stronger nucleophile** than H_2O

Hydrolysis of alkyl chlorides

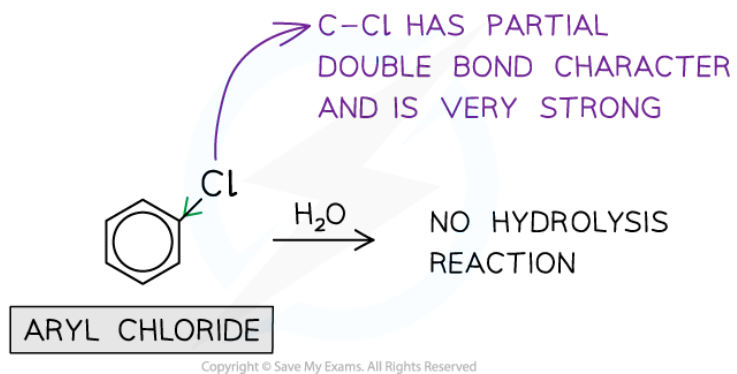


The hydrolysis of alkyl chlorides requires a strong nucleophile

Strength of C-Cl bond in aryl chlorides

- In aryl chlorides, the carbon atom bonded to the chlorine atom is part of the **delocalised π bonding system** of the benzene ring
- One of the lone pairs of electrons of the Cl atom **overlaps** with this **delocalised** system
- The C-Cl bond, therefore, has some **double-bond character** causing it to become **stronger**
- As a result, the C-Cl bond is difficult to break and **hydrolysis will not occur**

Hydrolysis of aryl chlorides



Due to the strong C-Cl bond in aryl chlorides, these compounds will not undergo hydrolysis